rate quantum devices. It helped the UCLA group to achieve a preparation and measurement error rate of about 0.03 per cent, lower than any other quantum technology to date.

Two-dimensional codes that currently exist require distillation, more precisely known as magic-state distillation. This is where the quantum processor sorts through the multiple computations and extracts the useful ones. This consumes a lot of computing hardware for just suppressing the errors.

Dr Brown of Sydney University had applied the power of three-dimensional code to a two-dimensional framework. The trick was to use time as the third dimension. In this research, he and his colleagues developed a decoder that identifies and corrects more errors than ever before, achieving a world record in error correction.

Silicon and electro tunnelling technology

Currently, regular qubits have to be kept at a temperature that is just fractions of a degree over absolute zero. In order to maintain this temperature, top of the line refrigeration technology is used. This costs millions of dollars and takes a large amount of space. A full-scale quantum computer using regular qubits would require an entire building for storing the cooling units. Recently, two groups, independently from Australia (Andrew Dzurak and Henry Yang from the University of New South Wales) and the Netherlands (Menno Veldhorst from the Delft University of Technology), have built quantum computers that operate at fifteen times higher temperature than before.

Instead of requiring dilution refrigerators that use isotopes helium-3 and helium-4, the system can be cooled using only helium-4. This is really cold (colder than space) and requires only thousands of dollars in refrigeration rather than millions. At higher temperatures, as described in the new research, control electronics could be placed right next to qubits on the same chip. This would reduce the costs of building quantum systems.

Qubit pairs are the fundamental units of quantum computing. Unlike a bit, however, they can manifest both states simultaneously, which is known as superposition.

The unit cell developed by Dzurak's team comprises two qubits confined in a pair of quantum dots embedded in silicon. The result, scaled up, can be manufactured using existing silicon chip factories and would operate without the need for multimillion-dollar cooling. It would also be easier to integrate with conventional silicon chips, which will be needed to control the quantum processor.

Every qubit pair added to the system increases the total heat generated and leads to errors. This is why current designs need to be kept close to absolute zero. The prospect of maintaining quantum computers with enough qubits at temperatures much colder than deep space is daunting, expensive, and pushes refrigeration technology to the limit. The South Wales team, however, has created an elegant solution to the problem by initialising and reading the qubit pairs using electrons tunneling between two quantum dots—this is again a quantum phenomenon.

Google is building a device that will usher in a new era for computing. It's a quantum computer designed to prove that machines exploiting exotic physics can outperform the world's top supercomputers. The company's goal, audaciously named Quantum Supremacy, is to build the first quantum computer capable of performing a task no classical computer can do. Google wants to claim the prize with just fifty-qubits.

Ion trap technology

Ion trap technology can be readily used to build a quantum computer by using individual atoms or ions. This is called an ion-based quantum computer. Kenneth Brown of Duke University, USA, says, "These basically have no memory error. Zeroing in on ion traps for quantum computing isn't new but it has not received the same notice that semiconductor-based superconducting approaches have."

Superconducting circuitry exploits the significant advantages of modern lithography and fabrication technologies. It can be integrated on a solid-state platform, and many qubits can simply be printed on a chip. However, these suffer from inhomogeneities and decoherence as no two superconducting qubits are the same, and their connectivity cannot be reconfigured without replacing the chip, or modifying the wires connecting them within a very low-temperature environment.

Trapped atomic ions, on the other hand, feature virtually identical qubits, and their wiring can be reconfigured by modifying externally applied electromagnetic fields. However, atomic qubit switching speeds are generally much slower than solid-state devices, and the development of engineering infrastructure for trapped ion quantum computers as well as mitigation of noise and decoherence from the applied control fields are just beginning.

India

India's Department of Science and Technology (DST) has initiated a programme called QuEST (Quantum-Enabled Science and Technology) to help lay the groundwork for building quantum computers in India. As a part of the programme, it will invest a sum of eighty billion rupees in a span of five years to facilitate research in this field.

India has joined the race for 'Quantum Supremacy' with DST for building the country's own fifty-qubit quantum computer in the next four to five years. DST has already started working on a draft for the mission.